

SHUMIKHA, Russian Federation

WMO: 286550

Lat: 55.233N

Lon: 63.317E

Elev: 178

StdP: 99.21

Time Zone: 5.00 (E05)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-30.1	-27.1	-32.7	0.2	-29.4	-29.9	0.2	-27.1	9.1	-5.6	8.2	-8.7	1.6	320	0.594

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.1	31.1	20.6	29.2	19.8	27.4	19.0	22.5	28.7	21.2	26.8	20.0	25.5	3.3	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.3	26.4	19.2	14.2	24.9	17.9	13.2	23.3	67.1	29.0	62.5	26.9	58.2	25.4	27.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.3	6.4	-33.5	34.0	3.4	1.8	-35.9	35.3	-37.9	36.4	-39.8	37.4	-42.3	38.7		
			-33.5	23.3	3.2	2.2	-35.9	24.9	-37.8	26.3	-39.6	27.5	-41.9	29.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	3.1	-15.6	-13.1	-5.6	5.1	12.7	17.9	19.1	17.5	11.3	3.8	-5.4	-12.1
	DBStd	13.54	6.37	6.59	4.95	5.52	5.12	4.60	3.75	4.47	4.55	5.04	6.97	7.14
	HDD10.0	3510	794	648	483	165	32	3	0	2	37	199	462	685
	HDD18.3	5747	1052	881	742	396	185	63	37	71	216	449	712	943
	CDD10.0	974	0	0	0	19	116	239	281	234	75	9	0	0
	CDD18.3	168	0	0	0	0	11	49	59	44	4	0	0	0
	CDH23.3	1744	0	0	0	8	164	521	555	446	49	0	0	0
	CDH26.7	539	0	0	0	1	39	169	168	152	11	0	0	0
Wind	WSAvg	2.9	2.9	3.0	3.2	3.4	3.2	2.9	2.5	2.5	2.7	3.0	3.1	3.0
Precipitation	PrecAvg	409	19	14	18	18	45	43	71	58	34	32	28	26
	PrecMax	584	35	28	40	41	116	86	220	150	84	83	62	47
	PrecMin	237	4	2	1	0	6	13	10	4	0	2	6	11
	PrecStd	97	10	7	12	11	34	19	48	40	21	19	14	10
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	2.5	9.8	24.2	30.0	32.8	33.6	33.2	28.3	19.4	8.2	2.4
		MCWB	-0.4	0.8	6.4	14.2	18.4	21.0	21.1	21.2	16.8	12.1	5.9	0.3
	2%	DB	-1.5	0.9	4.6	20.0	27.5	30.7	30.5	30.5	24.4	16.8	6.0	0.9
		MCWB	-2.5	-0.6	2.2	11.5	17.0	20.6	21.4	20.1	15.4	11.5	4.4	-0.4
	5%	DB	-3.5	-1.4	2.8	17.0	25.0	28.7	28.4	28.2	21.6	14.0	4.2	-0.7
		MCWB	-4.3	-2.6	0.9	10.4	15.7	19.6	20.5	19.5	15.2	9.8	2.6	-1.6
	10%	DB	-5.6	-3.7	1.5	14.2	22.3	26.4	26.4	25.7	18.7	11.2	2.6	-2.4
		MCWB	-6.4	-4.8	-0.1	8.8	14.4	18.6	19.4	18.7	13.8	8.1	1.4	-3.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-0.2	1.2	6.4	14.1	20.4	23.3	24.2	23.7	18.6	13.9	6.2	0.9
		MCDB	0.6	2.0	9.8	22.7	27.9	30.3	29.9	30.1	22.8	17.8	7.6	2.3
	2%	WB	-2.4	-0.6	2.5	12.2	18.2	21.9	22.6	21.9	16.8	11.7	4.5	-0.2
		MCDB	-1.5	0.8	4.2	18.5	25.7	28.6	28.5	28.1	22.3	15.9	5.9	0.7
	5%	WB	-4.3	-2.5	1.2	10.9	16.6	20.5	21.5	20.3	15.4	10.0	2.9	-1.6
		MCDB	-3.5	-1.4	2.6	16.1	23.4	26.8	26.6	26.0	21.0	13.7	4.0	-0.6
	10%	WB	-6.4	-4.7	0.0	9.2	15.1	19.3	20.2	18.9	14.0	8.4	1.5	-3.3
		MCDB	-5.7	-3.6	1.3	14.0	21.3	25.3	24.8	24.5	18.4	11.1	2.4	-2.5
Mean Daily Temperature Range	5% DB	MDBR	7.4	8.9	8.5	9.2	11.3	10.8	10.1	10.3	9.2	7.3	6.0	7.0
		MCDBR	8.1	8.5	8.3	13.6	15.2	14.0	13.3	14.3	13.2	11.8	6.3	7.1
		MCWBR	7.5	7.6	6.5	7.4	7.3	6.4	5.9	6.1	6.9	7.2	4.9	6.5
	5% WB	MCDBR	7.9	8.6	7.5	12.8	13.9	13.1	11.7	12.6	12.3	11.1	6.0	7.3
		MCWBR	7.5	7.8	6.1	7.5	7.8	7.0	6.3	6.8	7.0	7.1	5.1	6.8
Clear-Sky Solar Irradiance	taub	0.284	0.314	0.368	0.409	0.408	0.405	0.425	0.443	0.384	0.355	0.312	0.283	
	taud	2.063	2.038	1.968	2.121	2.222	2.303	2.248	2.163	2.292	2.241	2.154	2.037	
	Ebn at Noon	608	731	773	806	834	841	813	766	767	685	571	504	
	Edh at Noon	66	99	136	135	130	121	126	129	99	82	61	54	
All-Sky Solar Radiation	RadAvg	0.89	1.98	3.53	4.48	5.42	5.86	5.64	4.43	3.00	1.54	0.79	0.56	
	RadStd	0.07	0.08	0.22	0.45	0.51	0.56	0.44	0.44	0.25	0.23	0.10	0.07	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46)	Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page